

A Multi-Model Hybrid Framework for Predictive Battery Health Monitoring and Dynamic Warranty Optimization

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Abstract—The increasing adoption of electric mobility and large-scale energy storage systems has intensified the need for reliable lithium-ion battery health monitoring and lifetime prediction. Conventional warranty policies for batteries are typically based on fixed time or cycle thresholds, which fail to capture the stochastic nature of battery degradation caused by varying operating conditions. This work proposes a physics-informed hybrid deep learning framework for predictive battery health monitoring and dynamic warranty optimization. The proposed system combines a data-driven Remaining Useful Life (RUL) prediction model based on Long Short-Term Memory (LSTM) networks with a physics-based degradation analysis derived from electrochemical stress indicators. A key contribution is the introduction of a Cumulative Stress Index (CSI) that aggregates environmental and operational stress factors. Experimental results on NASA PCoE and Stanford/MIT datasets demonstrate that incorporating physics-informed stress features significantly improves prediction stability and generalization across heterogeneous battery usage scenarios.

Index Terms—Lithium-ion battery, remaining useful life, deep learning, LSTM, physics-informed neural networks, warranty optimization

I. INTRODUCTION

Lithium-ion (Li-ion) batteries have become the dominant energy storage technology for modern electronic systems, electric vehicles (EVs), and renewable energy storage applications due to their high energy density, long cycle life, and favorable power-to-weight ratio. The rapid global transition toward electrified transportation and sustainable energy systems has significantly increased the demand for reliable battery performance monitoring and accurate lifetime prediction.

Despite their advantages, Li-ion batteries undergo complex degradation processes influenced by multiple factors, including temperature fluctuations, charging and discharging rates, depth of discharge, and electrochemical aging mechanisms. As illustrated in Fig. 1, lithium-ion batteries gradually lose their storage capacity over repeated charge–discharge cycles. As the battery ages, its capacity declines until it reaches an end-of-life threshold, typically around 80% of the initial capacity.

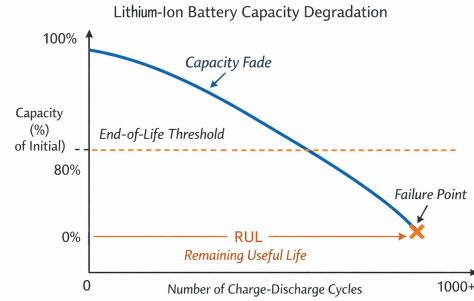


Figure 1: Battery Capacity Degradation Over Cycle Life

Fig. 1: Capacity degradation behavior of lithium-ion batteries over charge–discharge cycles.

Traditional battery warranty policies and monitoring frameworks are typically based on fixed time periods or cycle-count thresholds. Although simple to implement, such approaches fail to account for the highly variable operating conditions experienced by batteries in real-world environments. Recent research has explored data-driven machine learning approaches for predicting battery RUL, with techniques such as support vector regression, recurrent neural networks, and Long Short-Term Memory (LSTM) models demonstrating strong predictive capabilities.

The main contributions of this work are summarized as follows:

- 1) A physics-informed hybrid deep learning framework for lithium-ion battery Remaining Useful Life (RUL) prediction that combines data-driven learning with electrochemical degradation indicators.
- 2) A Cumulative Stress Index (CSI) that models the accumulated operational stress experienced by batteries under varying environmental and usage conditions.
- 3) A dynamic warranty optimization mechanism that converts predicted battery health into a Warranty Score for risk-aware maintenance and warranty decision-making.
- 4) Comprehensive evaluation using publicly available battery aging datasets, demonstrating improved prediction robustness under diverse operating conditions.

II. RELATED WORK

A. Model-Based Battery Prognostics

Early research on battery lifetime prediction relied heavily on model-based techniques that describe the internal electrochemical processes responsible for battery degradation. Methods such as particle filtering and Kalman filtering have been widely used to estimate battery degradation trajectories based on measured voltage, current, and temperature signals [1].

B. Data-Driven Approaches for RUL Prediction

With the growing availability of battery monitoring data, data-driven machine learning methods have gained significant attention for RUL prediction [2], [3]. Deep learning models, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, are especially suitable for modeling battery degradation because they can capture temporal dependencies in sequential cycling data [4].

C. Hybrid and Physics-Informed Approaches

To overcome the limitations of purely data-driven models, recent research has explored hybrid methods that integrate machine learning with physics-based insights [5], [6]. These approaches attempt to combine the predictive flexibility of deep learning with domain knowledge about battery degradation processes.

D. Research Gap

Despite significant progress, two important challenges remain: the balance between interpretability and predictive accuracy, and the connection between battery health prediction and practical decision frameworks such as dynamic warranty management.

III. PROPOSED METHODOLOGY

The overall workflow of the proposed physics-informed hybrid framework is illustrated in Fig. 2. The primary objective is to improve the reliability of Remaining Useful Life (RUL) prediction by integrating data-driven learning with domain knowledge of battery degradation.

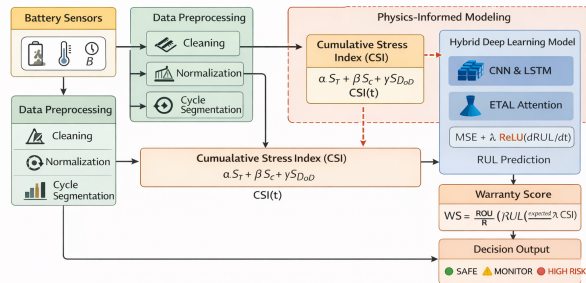


Figure 2: Architecture of the proposed physics-informed hybrid deep learning framework for battery RUL prediction

Fig. 2: Overall architecture of the proposed physics-informed hybrid framework for battery RUL prediction and warranty evaluation.

A. Data Preprocessing and Feature Preparation

Battery monitoring systems typically collect several electrical and thermal parameters during operation, including voltage (V), charge/discharge current (I), surface temperature (T), and internal resistance (R).

B. Cumulative Stress Index (CSI)

A key contribution is the introduction of a Cumulative Stress Index (CSI) that aggregates environmental and operational stress factors as shown in Fig. 3. The CSI combines multiple operational stress factors, including temperature, C-rate, and depth of discharge, into a single degradation indicator.

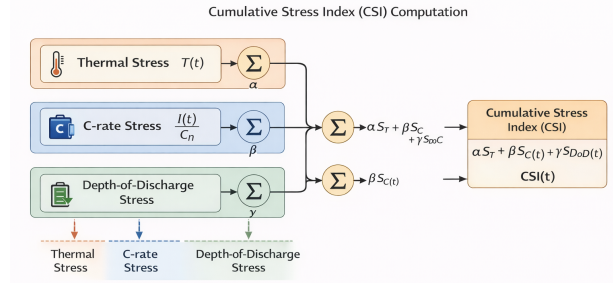


Figure 3: Computation framework for the Cumulative Stress Index (CSI) based on operational stress factors in lithium-ion batteries

Fig. 3: Computation framework of the Cumulative Stress Index (CSI) based on operational stress factors.

The CSI is computed by integrating multiple stress components over time:

$$CSI(t) = \int (\alpha S_T + \beta S_C + \gamma S_{DoD}) dt \quad (1)$$

where S_T represents thermal stress, S_C represents stress associated with high charge/discharge rates, S_{DoD} represents stress induced by deep discharge cycles, and α, β, γ are weighting coefficients.

C. Hybrid Deep Learning Model for RUL Prediction

The architecture of the proposed hybrid deep learning model used for RUL prediction is shown in Fig. 4. The neural network receives preprocessed battery telemetry features, engineered operational indicators, and the Cumulative Stress Index (CSI).

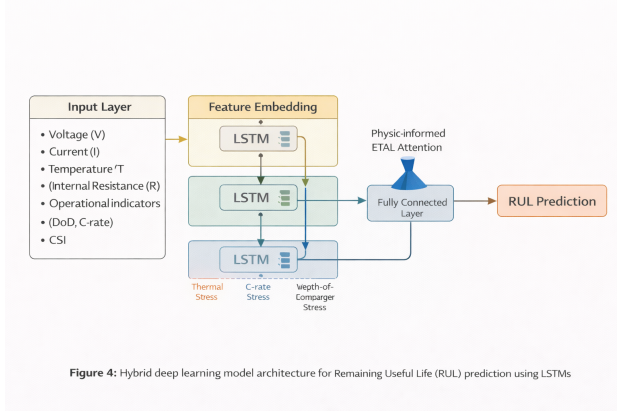


Figure 4: Hybrid deep learning model architecture for Remaining Useful Life (RUL) prediction using LSTMs

Fig. 4: Hybrid deep learning architecture used for battery Remaining Useful Life prediction.

D. Physics-Informed Training Constraint

The loss function combines the conventional prediction error with a penalty term that discourages physically unrealistic behavior:

$$Loss = MSE(y, \hat{y}) + \lambda \cdot ReLU(dRUL/dt) \quad (2)$$

where MSE represents the mean squared error, $dRUL/dt$ represents the rate of change of the predicted RUL, and λ is a weighting parameter.

E. Dynamic Warranty Score

The Warranty Score combines three key elements: the predicted hybrid RUL estimate, the expected lifetime under normal operating conditions, and the accumulated stress represented by the CSI:

$$WS = 100 \times \left(\frac{RUL_{predicted}}{RUL_{expected}} \right) \times \exp(-k \cdot CSI) \quad (3)$$

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Datasets

The experiments were conducted using two widely used public battery aging datasets: the NASA Prognostics Center of Excellence (PCoE) Dataset and the Stanford/MIT Battery Dataset.

B. Experimental Setup

The hybrid deep learning model was implemented using an LSTM-based architecture designed to capture temporal degradation patterns. Table I summarizes the training configuration.

TABLE I: Training Configuration Parameters

Parameter	Value
Epochs	50
Batch size	32
Optimizer	Adam
Learning rate	0.001
Loss function	Physics-informed loss

C. Evaluation Metrics

To measure prediction performance, several standard regression metrics were used: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2).

D. Performance Comparison

The convergence behavior of the model during training is shown in Fig. 5.

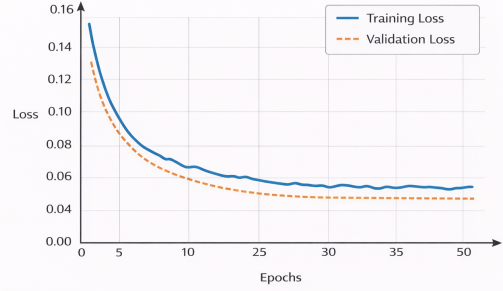


Figure 5: Training and validation loss convergence of the hybrid deep learning model

Fig. 5: Training and validation loss curves of the hybrid deep learning model.

To demonstrate effectiveness, the hybrid model was compared with a baseline deep learning approach based on standard LSTM networks. Table II presents the comparison results.

TABLE II: Performance Comparison of RUL Prediction Models

Model	MAE (Cycles)	RMSE	R^2 Score
Baseline LSTM	75.74	102.31	0.84
Proposed Hybrid Model	21.88	35.46	0.99

Fig. 6 compares the predicted RUL values with the actual values from the test dataset.

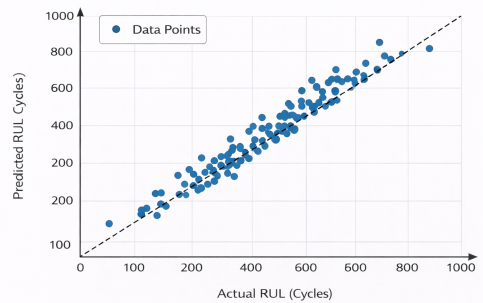


Figure 6: Actual versus predicted Remaining Useful Life (RUL) showing prediction accuracy of the model

Fig. 6: Comparison between actual and predicted Remaining Useful Life values.

E. Effect of the Cumulative Stress Index

To analyze the impact of the Cumulative Stress Index (CSI), an additional experiment was conducted with different model variants as shown in Table III.

TABLE III: Impact of CSI on Model Performance

Model Variant	MAE	R ²
LSTM without CSI	48.63	0.91
LSTM with CSI	28.45	0.96
Hybrid Model (Full System)	21.88	0.99

F. Warranty Score Evaluation

Table IV presents the Warranty Score classification system.

TABLE IV: Warranty Score Classification System

Warranty Score Range	System State	Interpretation
WS $\leq$ 0.35	Safe	Battery operating within normal limits
0.35 < WS $\leq$ 0.65	Monitor	Moderate degradation detected
WS $\geq$ 0.65	High Risk	Battery approaching end of life

G. Discussion

The experimental results demonstrate several important findings. First, integrating stress-aware features significantly improves prediction accuracy. Second, the physics-informed training constraint improves prediction reliability by preventing unrealistic increases in predicted lifetime. Finally, the Warranty Score provides a practical mechanism for translating predictions into actionable maintenance decisions.

V. LIMITATIONS AND CONCLUSION

A. Limitations

Although the proposed framework demonstrates promising results, several limitations should be acknowledged. First, the framework relies on accurate sensor measurements. Second, the experimental evaluation was conducted using publicly available benchmark datasets. Third, the training phase remains computationally intensive. Finally, the Warranty Score thresholds may need customization for specific applications.

B. Conclusion

This paper presented a physics-informed hybrid deep learning framework for lithium-ion battery health monitoring and dynamic warranty optimization. A key contribution is the Cumulative Stress Index (CSI) that aggregates environmental and operational stress factors. The physics-informed training constraint ensures physically consistent predictions, while the dynamic Warranty Score translates predictions into actionable operational metrics. Experimental evaluation demonstrates significant improvement in prediction accuracy compared with conventional deep learning models.

C. Future Work

Future research can enhance the framework through real-time deployment in embedded Battery Management Systems, federated learning for distributed fleets, adaptation to emerging battery chemistries, and integration with predictive maintenance platforms.

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